Big Mountain Resort Pricing Strategy Analysis

Using Data Analytics and Machine Learning

Problem Identification

- The resort recently added a new chair lift, increasing operating costs by **$1.54M** this season.
- Current ticket price: **$81**.
- Management is unsure whether this price reflects the resort's true value.
- They suspect they are **not fully capitalizing on their facilities**.

Recommendation

Increase ticket price gradually toward $89–$90

- The Random Forest model predicts Big Mountain's fair price is **$89–$90**.
- However, because the resort is already one of the more expensive options in Montana, increases should be **incremental and tested**.

Key Findings

1. Ticket price is most strongly influenced by:

   - **Vertical drop**
   - **Lift infrastructure** (especially fast quads)
   - **Number of Runs**
   - **Skiable Terrain**

2. Big Mountain scores well on these features, supporting a higher price.

3. Small operational changes can increase revenue, but closing too many runs sharply reduces value.

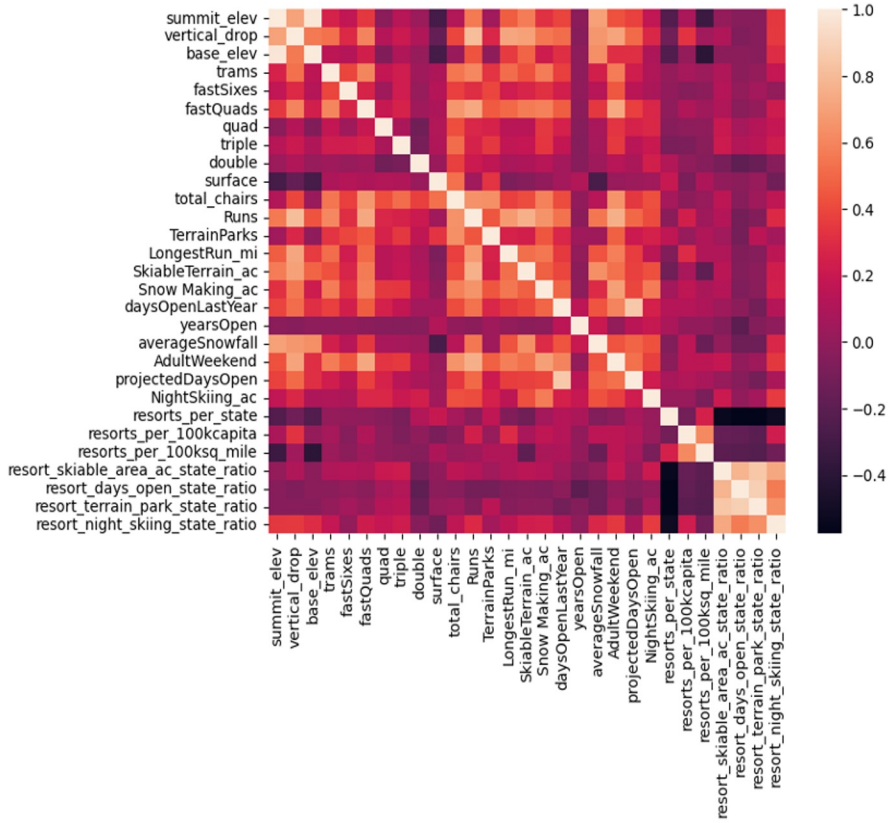
Data Overview

- Dataset: ~330 ski resorts
- Features included: elevation, runs, lifts, terrain, snowmaking, ticket prices
- Missing values removed; Adult Weekend ticket price chosen as target
- State-level data merged to enrich context

Exploratory Analysis

What drives ticket price?

- Strong positive correlations with:
  - Vertical drop
  - Number of runs
  - Total chairs
  - Fast quad lifts
  - Snowmaking area
- Larger resorts with better infrastructure charge more



Modeling Approach

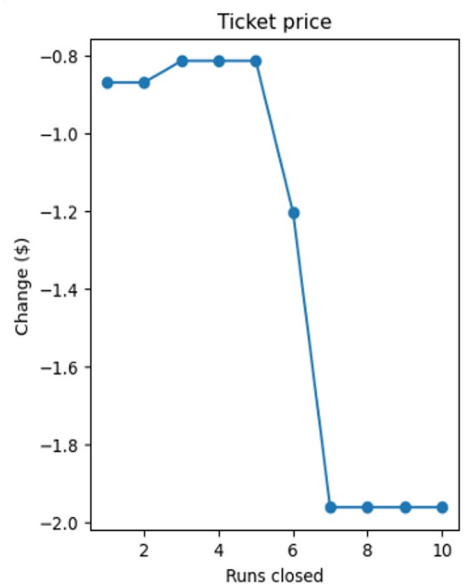
- Models tested:
  - Baseline (mean/median)
  - Linear Regression
  - Random Forest Regression
- Evaluation metric: **Mean Absolute Error (MAE)**
- Cross-validation: 5-fold

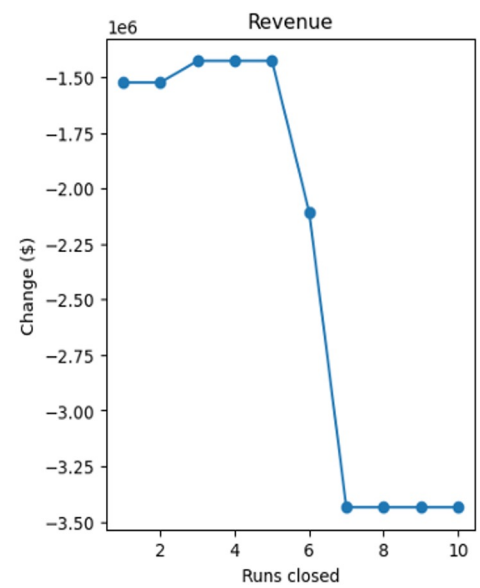
Model Performance

- Baseline MAE: ~18 (mean), ~10 (median)
- Linear Regression: improved but limited
- **Random Forest: MAE ≈ 9.12 → selected as final model**
- Feature importance confirms: runs, vertical drop, and lift infrastructure matter most

Scenario Modeling

- Adding a run or lift → small price increase → meaningful seasonal revenue gain
- Increasing snowmaking → minimal effect on price
- Closing 1–2 low-usage runs → minimal impact
- Closing 6+ runs → **sharp decline** in price and revenue





Final Recommendation

Move ticket price toward **$89–$90**, but:

- ❖ Increase gradually
- ❖ Monitor demand response
- ❖ Consider bundling or dynamic pricing

- Avoid closing more than 2 runs
- Invest in lift infrastructure where possible

Conclusion

- Ticket price is driven by resort size and lift quality.
- Big Mountain is positioned to charge more than $81.
- Data supports a higher price, but implementation must be strategic.

Future Work

- Incorporate:
  - Operating cost data
  - Customer demand elasticity
  - Competitor pricing behavior
- Build a profitability model
- Explore dynamic pricing strategies

Thank you.